AquaScope: Reliable Underwater Image Transmission on Mobile Devices

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Abstract

Underwater imaging is essential for scientific activities, such as ocean species monitoring. However, existing systems remain highly constrained by environmental challenges and often require specialized hardware, driving research into more accessible underwater imaging solutions. While recent acoustic-based communication systems support text messaging on mobile devices, their low data rates severely limit broader applications.

We present AquaScope, the first acoustic communication system capable of underwater image transmission on commodity mobile devices. To address the key challenges of underwater environments—limited bandwidth and high transmission errors—AquaScope employs and enhances generative image compression to improve compression efficiency, and integrates it with reliability-enhancement techniques at the physical layer to strengthen error resilience. We implemented AquaScope on the Android platform and demonstrated its feasibility for underwater image transmission. Experimental results show that AquaScope achieves a 36.7% improvement in visual quality while compressing a 256×256 color image into under 1k bits, outperforming existing systems across bandwidth-constrained and error-prone underwater conditions.

1 Introduction

Underwater imaging is of significant importance for scientific and operational activities, such as ocean species monitoring [57, 64], infrastructure inspection [58, 59] as illustrated in Fig. 1, and anomaly detection. For example, images enable accurate identification of marine organisms and support population and habitat assessments.

Despite its value, acquiring and transmitting underwater imagery *in situ* remains difficult due to fundamentally challenging underwater environments. Wireless technologies, such as cellular, Wi-Fi, or Bluetooth, are ineffective underwater because radio frequency (RF) signals suffer substantial attenuation in aquatic environments [15, 16]. While wired networks offer an alternative, they are often too cumbersome and impractical underwater. Existing research [4, 13, 38, 56, 78, 84] and industrial productions [24, 25, 72] addressing these limitations relies on specialized, sophisticated hardware, which is not accessible to non-expert users (§2.1).

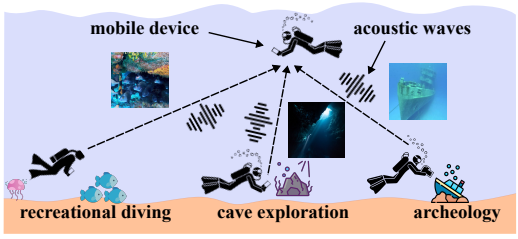


Figure 1: Conceptual graph of an underwater image transmission system for scuba divers.

The latest advancement in ubiquitous underwater communication utilizes acoustic signals to enable text messaging on mobile devices [15], as acoustic signals are known to propagate better in water than RF signals. Nevertheless, with its limited data rate—only 0.6 kbps beyond 10 meters—its practical applications are highly restricted.

In this study, we investigate the feasibility of *underwater image transmission on mobile devices*. Despite active research efforts in both industry and academia [4, 25, 49, 62, 72], implementing this technology faces two major challenges posed by underwater environments: (1) severely limited bandwidth, which renders conventional image codecs infeasible due to their inadequate compression efficiency, and (2) high transmission errors caused by varying channel conditions, which may substantially degrade reconstructed image quality or even prevent image decoding altogether (§2.2).

We present AquaScope, the first underwater communication system that enables image transmission on commodity mobile devices. AquaScope is also acoustic-based, but it is specifically designed to optimize bandwidth efficiency and enhance error resilience. At its core, AquaScope employs generative image compression [53, 70, 97], a recent technique that achieves significantly higher compression rates than traditional image codecs (e.g., JPEG [83]) and classic neural codecs [20, 48]. This generative model encodes an image into a sequence of tokens, each representing a small patch (e.g., 16×16 pixels) of the original image in a more *bandwidth-efficient* manner (§2.3). These tokens are then modulated by AquaScope into acoustic signals and transmitted underwater. If a token is corrupted or lost during transmission,

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the image decoder can leverage its generative capabilities to reconstruct image, thereby *mitigating transmission errors*.

Reaping the benefits of the above design in practice, however, comes with several challenges. First, generative compression models are typically pre-trained on large image datasets spanning diverse scenes. However, in the domain-specific underwater scenario, our analysis (Fig. 9) reveals that more than half of the tokens in the codebook are never used, leading to wasted bandwidth. Second, existing generative models assume that all encoder-generated tokens are received intact by the decoder. In reality, underwater transmission is prone to errors that can distort tokens. Third, the constrained computational and communication resources on commodity mobile devices necessitate efficient reliability-enhancing techniques at the physical (PHY) layer to correct errors in data symbols and ensure that tokens remain recoverable by the generative decoder after demodulation. We address these challenges using three key techniques.

Context-aware token clustering (§4.1). To improve bandwidth efficiency, we propose a token clustering approach tailored for underwater scenarios that reduces the effective codebook size in a generic generative compression model. Our method is motivated by the observation that *unused tokens lie close to frequently used ones in the embedding space*. We employ k-means clustering, where each cluster is represented by a new embedding computed as the weighted sum of its members. This strategy achieves a higher compression ratio by lowering the number of bits required per token while preserving perceptual image quality. We note that this token clustering technique can also be applied to other low-bandwidth communication settings (e.g., LoRaWAN [71]).

Error-resilient fine-tuning (§4.2). To enhance the error resilience of a generative compression model, we fine-tune its decoder on underwater datasets under simulated random token disturbances. While this approach resembles prior loss-resilient codecs proposed for video systems [19, 46], a fundamental difference in AquaScope is that underwater transmission errors distort data symbols directly at the PHY layer, making it impossible for the decoder to explicitly identify and mask corrupted (or lost) tokens, as done in prior methods. As a result, we must strategically perturb tokens during training, ensuring that the decoder learns to reconstruct images *without prior knowledge of corrupted tokens*.

Reliability enhancement at the PHY (§4.3, §4.4). Improving error resilience at the PHY layer is equally important, as it minimizes the number of corrupted tokens passed to the generative image decoder after demodulation, thus helping to preserve the reconstructed image's quality. To achieve this, we first insert multiple training symbols into the data packet. The number of training symbols is determined through offline profiling, selecting a value that balances packet overhead against the symbol error rate (SER). These training symbols enable the receiver to perform time equalization for accurate channel estimation, improving the recovery of data symbols. Next, to address symbol time offsets caused by the underwater device movement, we propose an effective time synchronization algorithm based on two key observations specific to the underwater communication environment: *the relative speed between the sender and receiver is inherently limited underwater*, and their *distance changes gradually without sudden shifts*. Leveraging these physical principles, AquaScope filters out unrealistic detected start timestamps and smooths the detection results, thereby obtaining accurate start timestamps for symbols.

We developed a prototype of AquaScope on the Android platform and conducted extensive real-world experiments across diverse aquatic settings: dock, bridge, lake bank, pedal boat, and lake beach. Our results show that: (1) AquaScope compresses a 256×256 color image into less than 1k bits while preserving high fidelity, yielding an average Learned Perceptual Image Patch Similarity (LPIPS) [99] score of 0.24. In the presence of transmission errors, it delivers a 36.7% improvement over existing systems; (2) AquaScope reliably transmits 256×256 color images under 9s over distances of up to 20m; (3) AquaScope maintains robust performance across various depths, orientations, distances, and mobility conditions, reducing the bit error rate (BER) from 19% to below 2% on average compared with state-of-the-art systems. We discuss the limitations of AquaScope in §7. Our key contributions are as follows:[1]

- We design and implement AquaScope, the first underwater acoustic system that enables reliable image transmission between mobile devices.
- AquaScope employs generative image compression and enhances its bandwidth efficiency and error resilience for underwater communication (through context-aware token clustering and error-resilient fine-tuning).
- AquaScope implements reliability-enhancing techniques at the PHY layer to mitigate the impact of underwater transmission errors.
- AquaScope demonstrates the feasibility of image transmission in bandwidth-limited, error-prone underwater environments on resource-constrained mobile devices.

2 Background and Motivations

2.1 Underwater Image Transmission Systems

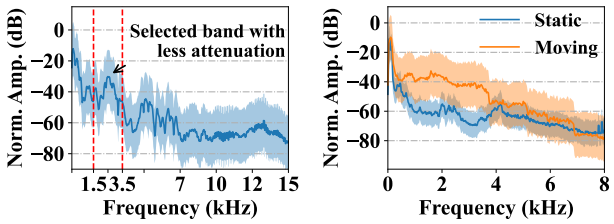
Underwater imaging has garnered increasing attention across both academia and industry. Many systems primarily rely on

[1]The demo and code are available at: https://chutoutian.github.io/aquascope.github.io/

offline recording, where cameras capture images or videos that are later retrieved and transferred to end devices (e.g., smartphones or PCs) via wired or short-range wireless links (e.g., Bluetooth [25]). More recently, the field has progressed toward real-time underwater scene monitoring and transmission [4, 7, 8, 17, 25, 56, 72, 96, 100]. Most existing underwater communication systems rely on acoustic signals, as they exhibit significantly lower attenuation in water compared to radio-frequency (RF) signals (e.g., Wi-Fi and Bluetooth) and optical signals [13, 47]. In contrast to terrestrial wireless networks, which benefit from mature and widely adopted standards such as 4G/5G and Wi-Fi, underwater communication is typically realized through *custom-designed protocols* and *hardware*, tailored for specific deployment scenarios.

Specialized acoustic hardware. Some of researchers [4, 56] leverage piezo-acoustic backscatter to enable battery-free underwater image sensing. A cylindrical piezoelectric transducer converts the harvested energy at the tag into acoustic waves, which are captured by a hydrophone at the receiver and subsequently decoded into images. Similarly, Anjum et al. [8] utilize customized transducer and hydrophone receivers for underwater image transmission, combined with OFDM (Orthogonal Frequency Division Multiplexing)-based acoustic modulation. In parallel, several industrial products have been introduced [24, 25, 72]. For instance, Sercel [72] integrates sophisticated underwater acoustic modems with hydrophone arrays. While such sophisticated hardware devices offer high transmit power and wide frequency bandwidth, enabling long-range, high-rate underwater links, they remain expensive and specialized, and thus are not easily accessible for non-expert users. *In contrast, AquaScope aims to provide an end-to-end underwater image transmission solution that runs entirely on commodity mobile devices, without relying on specialized acoustic hardware.*

Underwater image compression. Prior work has explored improving image compression for underwater transmission across both classical and learning-based designs [7, 8, 17, 45, 49, 96, 100]. For example, SeaScan [56] applies conventional JPEG compression [83]. Beyond off-the-shelf codecs, Anjum et al. [7, 8] propose a joint source–channel coding framework based on convolutional neural networks, training the encoder–decoder end-to-end over a theoretical channel model to minimize reconstruction distortion. More recent work explores leveraging large vision language models [9, 60] to transmit compact semantic descriptions or a small set of salient objects, and then reconstruct the scene at the receiver [15, 17, 45, 100]. However, most existing approaches are validated primarily in simulation [7, 8, 17, 45, 49, 96, 100]. As a result, *their real-world effectiveness, particularly on resource-constrained mobile platforms, remains underexplored.*



(a) Channel response (b) Noise

Figure 2: Characterizing underwater communication: bandwidth is limited, and the channel is noisy.

2.2 Challenges on Mobile Devices

In this work, our goal is to push the boundaries of underwater communication on ubiquitous mobile devices by exploring the feasibility of image transmission. We implement AquaApp [15], the state-of-the-art acoustic communication system that only supports text messaging on mobile devices, on Samsung S21 mobile phones and extend it to support image transmission. Specifically, images are encoded into byte streams using different codecs, including JPEG [83], PNG [82], and classic neural codecs [20]. These encoded bytes are then transmitted using AquaApp's three-step OFDM protocol. The process begins with the sender transmitting a preamble to the receiver. Based on this preamble, the receiver identifies and replies with high SNR (signal-to-noise ratio) subcarriers. The sender then encodes the payload on the selected subcarriers and transmits the data. The phones were sealed in waterproof pouches and submerged for underwater experiments.

Limited bandwidth. Transmitting images underwater using commodity mobile phones is fundamentally constrained by the acoustic front end and the channel. First, a phone's built-in speaker and microphone support only a narrow effective audible band, and this usable band is further reduced by the frequency-selective underwater channel. Fig. 2a plots the channel frequency response (normalized amplitude in dB) where the shaded region indicates the variation. We observe a pronounced roll-off beyond roughly 3.5 kHz, implying that tones above this frequency are rapidly attenuated and therefore unreliable for communication [15, 91]. Second, underwater ambient noise such as air bubbles, water flow, and sounds produced by the waterproof bag often overlaps with the same audible spectrum used by phone acoustics, introducing additional interference. Fig. 2b shows the normalized ambient noise spectrum over 0–8 kHz, with especially strong noise concentrated in 0–1 kHz. Taken together, we restrict our transmission band to 1.5–3.5 kHz for image delivery.

Image codecs. We first show that existing codecs exhibit *low compression efficiency* and are *vulnerable to transmission errors*. Fig. 3 illustrates the transmission latency for a 256×256

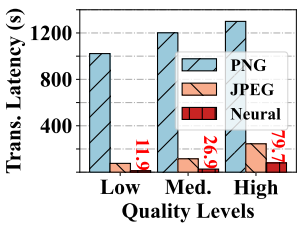


Figure 3: Transmission latency of images compressed with different codecs.

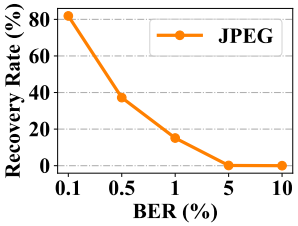


Figure 4: Recovery rate of JPEG (PNG and classic neural codecs fail to recover images).

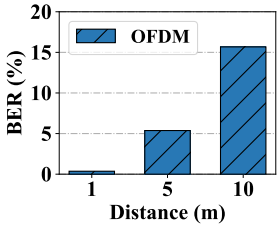


Figure 5: BER *vs.* distance.

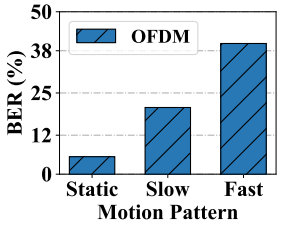


Figure 6: BER *vs.* mobility.

image at various compression levels. Since AquaApp's maximum effective data rate is only 0.6 kbps [15], it takes at least 80s, 1000s, 12s to transmit images encoded with JPEG (low quality level of 30), PNG (scaled to 30% of the original), and neural codecs (low quality level of 1), respectively, *assuming* no transmission errors. Note that low-quality settings introduce noticeable artifacts and blurring, as shown in Fig. 16, significantly reducing image clarity. This is because these codecs prioritize the reconstruction of pixel-level details, which proves inefficient for underwater transmission.

Meanwhile, errors are inevitable over extended transmission times due to channel variations. Existing image codecs are particularly vulnerable to transmission errors due to entropy coding [34, 86]. If a received packet is corrupted or incomplete, the image decoding process may fail. To quantify this, we define the *recovery rate* as the fraction of successfully decoded images among transmitted ones. We simulate lossy channels with varying BERs and compute the recovery rate over 500 images. The results, shown in Fig. 4, indicate that more than half of JPEG images fail to decode when the BER exceeds 0.5%. PNG and classic neural codec [19, 20] use sequential entropy coding, where bit errors propagate through the entire stream and prevent recovery. Therefore, the compressed images are unrecoverable even at a BER of 0.1%, so their results are omitted.

Transmission protocol. Then, we demonstrate the limitations of the OFDM protocol employed by AquaApp. Fig. 5 and Fig. 6 illustrate the BER under varying distance and mobility conditions, respectively. Even in the least demanding scenario, we observe that the BER reaches 0.36%, corresponding to a recovery rate of only about 60% (Fig. 4). Furthermore, the BER increases rapidly when the distance exceeds 5 meters, or the device is in motion, reaching up to 40% and rendering underwater image transmission infeasible.

Since AquaApp is designed for small packet transmissions (*i.e.,* text messages), its transmission protocol struggles with channel dynamics, extended transmission ranges, and mobility. The decline in SNR with increasing transmission distance leads to high demodulation errors, even with carefully selected subcarriers. Moreover, device movement can alter channel conditions and cause previously selected subcarriers to degrade in quality over the course of transmission. Addressing this requires frequent subcarrier reselection, which is time-consuming (e.g., ~3 seconds per selection).

2.3 New Opportunity: Generative Compression

Given the harsh conditions of underwater communication, an effective image compression scheme must reduce the image size substantially while achieving robustness against transmission errors, posing a challenge for existing image codecs [20, 48, 83]. However, recent advancements in deep generative models [28, 53, 97] offer a promising alternative—*generative compression*.

Fig. 7 depicts the workflow of generative compression. The encoder transforms a raw image into a sequence of *tokens*, which represent *indices*[2] of embeddings (*i.e.,* feature vectors) in a pretrained codebook. Unlike previous neural codecs [20] that encode images as a large set of feature values, this token-based method provides a more bandwidth-efficient representation of semantic information [53, 97]. After the image tokens are transmitted over the underwater channel, the decoder maps them to their corresponding embeddings and reconstructs the original image.

Generative compression offers an attractive solution for severely bandwidth-limited environments with "graceful degradation" property [70], where the reconstructed image quality gradually declines as bandwidth decreases or transmission errors occur. More importantly, a growing body of research work [53, 55] has demonstrated that token-based generative compression delivers superior perceptual quality at low bitrates. As we will show later, generative compression operates at bandwidth levels beyond the reach of conventional methods and reduces transmission latency by half compared with classic neural codecs. Moreover, even in the presence of transmission errors, the received image remains both decodable and semantically meaningful.

Summary:

- Existing ubiquitous underwater communication systems, designed for small packet transmissions, are inadequate for handling image data.
- Traditional image codecs face significant challenges in underwater environments due to limited bandwidth and vulnerability to transmission errors.

[2]The terms "tokens" and "indices" are commonly used interchangeably.

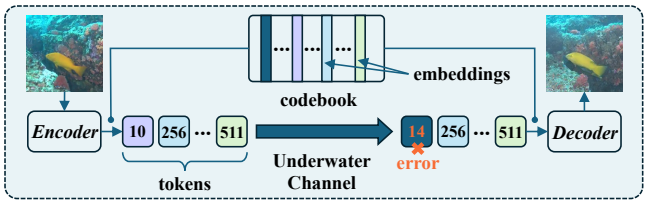


Figure 7: Overview of generative image compression.

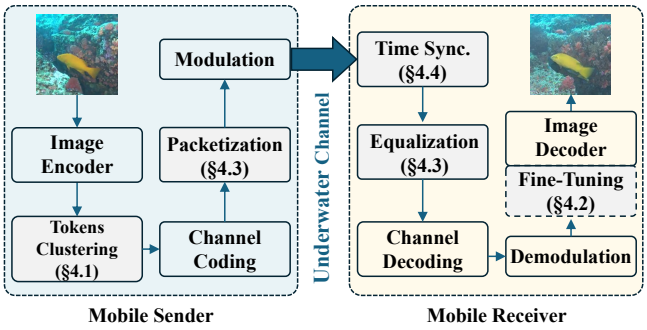


Figure 8: AquaScope overview.

- Generative image compression presents a promising solution for reliable underwater image transmission.

3 System Overview

Fig. 8 provides an overview of AquaScope. The image encoder and decoder are jointly trained offline on underwater datasets with simulated transmission errors (§4.2). During operation, the mobile sender compresses a raw image into tokens (i.e., indices of embeddings in a codebook) and converts them into a more compact bitstream using "context-aware token clustering" (§4.1). It then encodes the bits into data symbols via channel coding and encapsulates them into a packet at the PHY layer (§4.3). This packet includes a leading preamble followed by multiple symbol groups. Next, the sender modulates the packet data into an acoustic signal for transmission through the underwater channel to the receiver.

Upon reception, the mobile receiver performs preamble detection and time synchronization (§4.4) to locate symbol groups in the received signal. It then compensates for channel distortions through equalization (§4.3), demodulates the signal into data symbols, reconstructs the bitstream, and ultimately generates a high-fidelity underwater image.

4 AquaScope Design

In this section, we introduce three key designs of AquaScope: context-aware tokens clustering, error-resilient fine-tuning, reliability enhancement at the PHY layer including packetization and smoothed & bounded time synchronization.

4.1 Context-aware Tokens Clustering

Generative image compression models encode an image with M tokens, each selected from a codebook containing

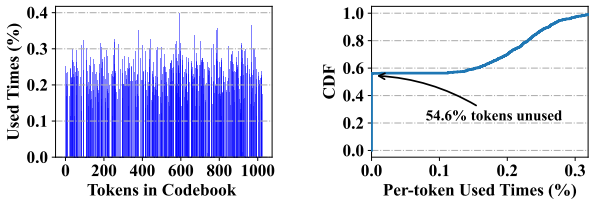


Figure 9: Tokens in codebook are underutilized: 54.6% of tokens remain unused in underwater image compression.

K possible tokens. This results in a compressed image size of $M \times \log_2 K$ bits. Designed to reconstruct and generate images with high fidelity across a wide range of scenes (e.g., ImageNet-1000 with 1,000 object classes), these generative models often conservatively adopt large values for M and K to preserve visual details.

In specialized domains such as underwater imaging, many tokens in the codebook are rarely or never used, leading to inefficient bandwidth usage. To quantify this redundancy, we analyze token utilization when compressing images from underwater datasets. As shown in Fig.9, 519 out of 1024 tokens in VQGAN's codebook [23] remain unused, highlighting the opportunity to optimize representation and improve compression efficiency. To address this, we propose a context-aware clustering approach that merges embeddings based on their Euclidean distance, motivated by the observation that the utilized tokens are approximately evenly distributed across the codebook, as illustrated in Fig. 9. Specifically, we adopt k-means clustering, where each cluster shares a new embedding computed as the weighted sum of its members, with weights proportional to their usage frequency. This process produces a more compact codebook and reduces the bit width of indices. For example, reducing a 1024-entry codebook to 64 entries lowers the index width from 10 bits to 6 bits, saving 40% of the data size while preserving semantic similarity.

4.2 Error-resilient Fine-tuning

Generative image compression models are commonly pre-trained on large-scale image datasets (e.g., ImageNet [5]) to achieve generalization, with the training process assuming that all tokens produced by the encoder are received intact by the decoder. However, in underwater environments, transmission errors are inevitable, often resulting in corrupted or lost tokens. Meanwhile, best practices suggest that fine-tuning a pre-trained generative model for a specific target scenario enhances performance. Therefore, we fine-tune the image decoder on underwater datasets with simulated transmission errors, as detailed below.

We denote the encoder and decoder by f_θ and g_ϕ, respectively, with θ and ϕ representing their trainable weights. As illustrated in Fig. 10, the encoder f_θ typically consists of convolutional layers that extract features and downsample a raw

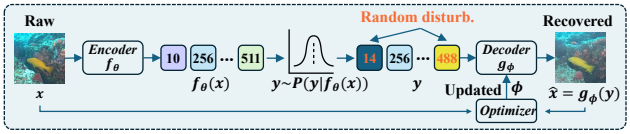


Figure 10: Fine-tuning generative image compression for underwater communication.

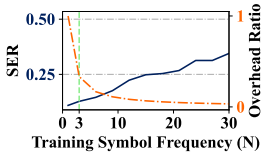


Figure 11: Symbol error rate (SER) *vs.* overhead ratio.

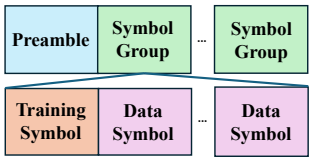


Figure 12: Image packetization.

image into a compact latent representation which is then converted into a sequence of indices, denoted by $f_\theta(\mathbf{x})$. Then, we randomly sample from these indices and introduce disturbance. The resulting sequence $\mathbf{y}$ is fed to the decoder, which uses upsampling and convolutional layers to reconstruct the image as $\hat{\mathbf{x}} = g_\phi(\mathbf{y})$. During training, we optimize the encoder and decoder parameters by minimizing a composite loss function that combines pixel-level reconstruction loss, perceptual loss, and GAN loss, following standard practices in the image reconstruction model training [23].

The random disturbance is introduced as follows. Given M tokens, we randomly perturb m of them (by replacing each with an alternative token from the codebook), where m is randomly selected from $[1, p]$ ($p \leq M$).[3] Following the curriculum learning strategy [10], we progressively increase p during training, allowing the model to learn with increasingly challenging perturbations for improved performance. We cap p at 25% of M to prevent excessive perturbation, which could eliminate the useful information in tokens and hinder the model's ability to learn. This 25% threshold is based on preliminary real-world experiments on error rates.

A key difference between AquaScope and prior loss-resilient streaming frameworks [19, 46] is that they assume the decoder can detect which indices are lost during transmission. This assumption holds in terrestrial network stacks, where indices are packetized and labeled with sequence numbers [12]. Thus, the missing packets, and hence missing indices, are explicit. In the underwater environment, enabling comparable loss visibility would require stronger FEC, incurring substantial overhead. While providing explicit masked-token information would help the decoder focus on corrupted positions, generative models like VQGAN [23] and TiTok [98] can still achieve strong recovery because their redundant codebooks and large-scale training data encode powerful natural image priors as illustrated in §6.2.

4.3 Packetization

To balance the throughput and reliability, AquaScope transmits the full set of indices as a single PHY-layer payload. Recall from §2.2 that underwater image transmission can take tens of seconds due to limited bandwidth, and is highly susceptible to transmission errors caused by channel variations at the PHY layer. To address these challenges, we insert a training symbol into the image packet after every N data symbols. This allows for more frequent channel equalization to correct symbol distortion introduced by the channel, and facilitates time synchronization to compensate for device movement. Taken together, more training symbols help reduce the symbol error rate (SER).

However, as shown in Fig. 11, increasing the number of training symbols (*i.e.*, reducing N) also raises the overhead ratio, leading to longer transmission and signal processing times. To balance this trade-off, we set $N = 3$, which ensures that the duration of each symbol group[4] remains within the channel coherence time underwater [93].

In traditional RF-based wireless systems, multiple training symbols are commonly used in OFDM [3]. However, as discussed in §2.2, the OFDM protocol encounters significant challenges in underwater image transmission, primarily due to substantial SNR drops in initially selected subcarriers during long mobile transmissions. Consequently, adding multiple training symbols for equalization becomes ineffective, as equalization fails when the SNR is too low. Instead, we adopt the CSS (chirp spread spectrum) technique [39, 75, 76, 88, 89] to convert image packets into chirp signals as it offers a balance between robustness and transmission speed. We use a preamble to enable packet detection and timestamping, which is critical at low SNR in underwater environments. Following [15], the chosen preamble offers strong auto-correlation and low false positives, ensuring robust detection against ambient noise.

In summary, each image packet consists of a single preamble, followed by multiple symbol groups, where each symbol group includes one training symbol and three data symbols as shown in Fig. 12.

4.4 Smoothed & Bounded Time Synchronization

The receiver's microphone continuously records audio and monitors for packet arrival using the detection algorithm in [15]. To identify packets, we perform cross-correlation between the received signal and the preamble. Once a correlation peak is found, an image packet is detected, and the receiver captures sound for a fixed duration (slightly longer than the packet length) as raw received data. Nevertheless,

[3] Bit errors can be introduced directly at the PHY layer. Because the acoustic communication involves a complex signal-processing pipeline (see in §5), such perturbations manifest at the application level as effectively random corruption of token indices.

[4] Given a symbol time T_s of 16 ms (as explained in Sec. 5), the duration of each symbol group is $(N + 1) \times T_s = 64$ ms.

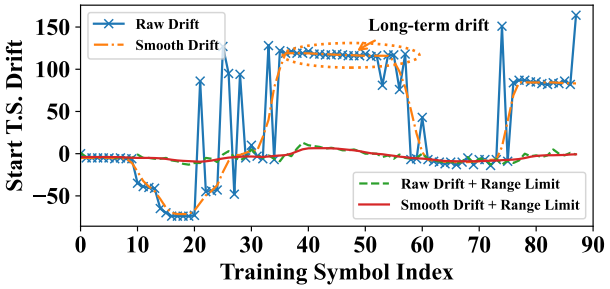


Figure 13: Start timestamp drift in training symbols.

due to the movement of mobile devices, symbol time offset is inevitable. To correct this offset, the receiver identifies the start timestamp of each training symbol via cross-correlation and adjusts the timestamps of the following data symbols. However, directly applying cross-correlation for symbol timing synchronization is error-prone due to channel fading, multipath effects, and ambient noise.

Therefore, to enhance the robustness of time synchronization, we leverage two insights related to physical principles. First, the relative speed between the sender and receiver during a specific activity (e.g., scuba diving) is naturally constrained, *i.e.,* the difference in start timestamps between two consecutive training symbols is bounded:

$$|t^{i-1}_{tsym} + N_{sgroup} - t^{i}_{tsym}| \leq \Delta, \tag{1}$$

where t^{i}_{tsym} denotes the start index of the i-th training symbol, and N_{sgroup} represents the length of the sending symbol group in samples. In our setup, $N_{sgroup} = (N+1)\times N_s = 3072$, where N_s is the number of samples per symbol. We set $\Delta = 40$ empirically based on preliminary experiments, taking into account the maximum relative movement speed between the sender and receiver, as well as orientation changes. This bounded constraint allows us to efficiently search for the start time of the next training symbol within a limited range, thereby improving synchronization accuracy and computational efficiency on mobile devices.

Second, the distance between the sender and receiver should change gradually, *i.e.,* the drift in the start timestamp of each training symbol should also change smoothly. We calculate the drift as the difference between the detected start timestamp of the training symbol and its expected start timestamp ($i\times N_{group}$) when there is no movement. If the drift changes drastically, resulting in excessive jitter as shown by the blue line in Fig. 13, we can infer that an error has occurred in symbol time synchronization.

In Fig. 13, the x-axis represents the index of the training symbol, and the y-axis shows the start timestamp drift. We collected this data by submerging two phones underwater from two boats and driving the boats in random directions and at varying distances. *Raw Drift* does not address jitter specifically, leading to noisy and inaccurate predictions due to imperfections in the cross-correlation results. Leveraging our second insight, we apply a moving average [35] to smooth *Raw Drift* and eliminate spikes, as shown by *Smooth Drift*. While this method effectively corrects short-term drift (and thus identifies the correct start timestamp quickly), it fails to handle long-term drift (where the timestamps shift beyond the recoverable range). To address this, we further constrain drift within a limited range based on our first insight. This is denoted as *Smooth Drift + Range Limit* and used to recalculate the start timestamp of each training symbol. Later, we demonstrate that our proposed drift smoothing and bounding approach significantly reduces SERs.

After compensating for the influence of movement and smoothing out detection errors, we utilize the correct start timestamps of two consecutive training symbols—obtained from the previous symbol time synchronization—to segment the symbol group and extract the data symbols. Finally, we apply linear interpolation to the data symbols to mitigate the Doppler effect.

5 Implementation

5.1 Generative Model

Dataset and model training. We curate two non-overlapping datasets for model training and testing. The training dataset comprises 78,420 images sourced from public underwater datasets [6, 26, 27, 32, 36, 37, 44, 54] and 135,000 general images sampled from ImageNet [5]. For testing, we extract 1,216 underwater images from five scuba diving videos [30, 41, 42, 77, 80], captured in various diving locations and conditions.
Model training and deployment. Building on the open-source code from [15], we implement two Android versions of AquaScope with different generative compression models. The first uses CNN-based encoding and decoding following VQGAN [23], where each index has 6 bits and a 256×256 image is mapped to 256 indices. The second adopts a Transformer-based design following TiTok [98], using 12-bit indices and compressing a 256×256 image into 64 indices. Note that the models are optimized by the context-aware tokens clustering. We ultimately use TiTok in our system due to its superior performance. We initialize our model using a pre-trained TiTok model [97] and fine-tune it for 112 epochs on our training dataset, followed by 60 epochs of error-resilient fine-tuning. These two stages require 11 and 14 hours of training time on 4×A100 GPUs, respectively.

The neural network models, initially in PyTorch format, are converted into Android-compatible formats and optimized for mobile devices using PyTorch [1] libraries. With a total size of under 1 GB, these models can be loaded and executed on modern smartphones, such as the Samsung S21 [69] used in our study.

5.2 Transmitter

The transmitter performs modulation and channel coding, converting serialized image bits from the image encoder into time-series acoustic signals. These signals are then broadcast into the water via the mobile device's speaker. To ensure a high SNR for acoustic signals, we select the 1.5–3.5 kHz frequency range for image transmission, based on the measurement study presented in Fig. 2a.

Modulation. The CSS modulation technique encodes each symbol as a chirp signal that spans the entire bandwidth [40, 89]. The chirp generation process is governed by two key parameters: the spreading factor (SF) and the bandwidth (BW). To balance robustness and data rate, we set $SF = 5$, achieving a data rate of 312 bps with $BW = 2$ kHz in underwater environments. Consequently, the symbol time is $T_s = 2^{SF}/BW = 16$ ms. After modulation, the mobile device transmits the acoustic signal at a sampling rate of $f_s = 48$ kHz, generating $N_s = 768$ samples per symbol.

Channel coding. To enhance robustness and facilitate error recovery, we employ encoding techniques commonly used in chirp-based communication systems [18, 71, 89]:

- *Hamming coding.* We first apply Hamming encoding [31] with a 4/7 code rate, encoding every 4 data bits with 3 parity check bits to form a 7-bit codeword.
- *Diagonal interleaving.* We group every $SF = 5$ codewords into 7 symbols, distributing each bit of a codeword diagonally across 7 different symbols [89]. This approach aligns with the fact that a (7, 4) Hamming code can correct up to 1-bit error per codeword.
- *Gray coding.* We use Gray coding [22] to assign binary values to symbols so that adjacent symbols differ by only one bit. This minimizes the impact of demodulation errors and improves the effectiveness of Hamming coding in detecting and correcting bit errors.

5.3 Receiver

The receiver performs several key functions, including packet detection, time synchronization, equalization, demodulation, and decoding, to extract data bytes from the received raw signal. These data bytes are subsequently forwarded to the image decoder for image reconstruction.

Equalization. We apply MMSE-based time-domain equalization [63] to each symbol group. The equalization coefficients are estimated using each training symbol and then applied to equalize the subsequent data symbols within the same group. To balance performance and computational complexity, we configure the equalizer with a tap size of 240 samples and an offset of 80 samples. Furthermore, the time-domain equalization process helps mitigate symbol time offsets during convolution operations.

Demodulation. After extracting each data symbol from the received packet, we apply a standard de-chirp process to demodulate the symbol, following the approach in [89].

Channel decoding. The channel decoder applies the reverse process of the channel encoder, incorporating Gray decoding, diagonal de-interleaving, and Hamming decoding. Once the data bytes are successfully decoded, they are passed to the image decoder for final image reconstruction.

6 Evaluation

6.1 Experimental Setup

To thoroughly evaluate our system, we conducted a series of tests in various environments using customized setups.

Experimental environments. We tested our system across different environments to assess the impact of distance, depth, orientation, mobility, and any other environmental factors on its performance. The environments considered are listed below and the corresponding images are shown in Fig. 14.

- *Fishing dock:* A quiet environment with limited space for setup, suitable for short-range experiments.
- *Bridge:* A realistic and quiet environment with ample space (a measurement distance of up to 20 m and an average water depth of 3 m), making it the primary test site.
- *Lake bank:* A quiet site that allows for easy control of the phone's orientation; bank side has a reflective surface.
- *Pedal boat:* A busy and noisy location with kayaks and pedal boats passing by; may simulate realistic underwater movement patterns.
- *Lake beach:* A site for testing underwater image transmission between two swimmers where they swim and tread water in random directions and speeds with mobile phones held in hand to mimic the real diving scenario.

As shown in Fig. 14, when operating far from water, we secure the phone pouch to a tape measure hook with a climbing lock. A barbell weight, attached to the same hook with a 0.5m rope, ensures the phone stays submerged. The tape measure works as both a rope and a depth gauge. The phone in the pouch floats vertically in the water, maintaining an upright orientation when it is static. When operating close to water, we attach the phone in its pouch to a three-meter-long selfie stick using a 0.5m rope, causing the phone's depth and orientation to vary randomly during the test.

The choice of the waterproof tool is essential for transmission quality and phone safety. After evaluating several options, as shown in Fig. 14, we choose a pouch for its minimal signal attenuation and reliable waterproofing (rated to withstand 30 hours at a depth of 100 feet) [79].

Baseline and metrics. Note that there is no existing mobile systems that support underwater image transmission. We construct five baseline schemes based on the existing image codecs and transmission protocols.



Figure 14: Experimental environments from left to right: fishing dock, bridge, lake bank, pedal boat, and lake beach. The remaining images show a prototype of AquaScope and the mounting setup used to secure phones to the measurement tool when close to and when far from the water.

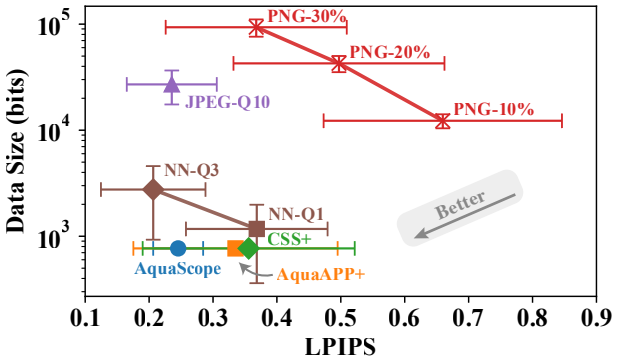


Figure 15: End-to-end performance comparison between AquaScope and baseline schemes.

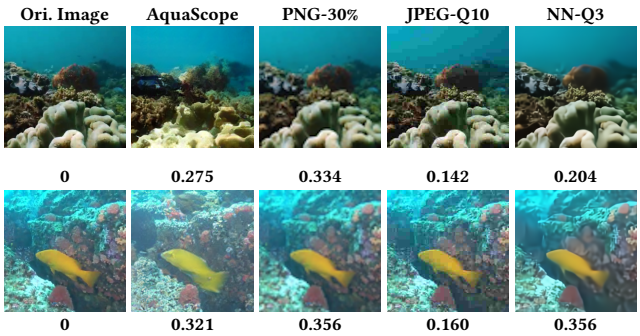


Figure 16: Samples of recovered images by AquaScope and baseline codecs. For each reconstructed image, the caption displays LPIPS. The baseline codecs are evaluated in error-free cases, while AquaScope is evaluated under 14% and 9% IERs, respectively.

- *AquaApp+ [15]:* AquaApp is originally designed for text transmission using a three-step OFDM protocol. Specifically, the sender transmits a preamble, the receiver responds with high-SNR subcarrier information, and the sender then encodes and sends the payload using those subcarriers. The redundancy of OFDM is set to 50%, and BPSK [65] modulation is adopted. This setup provides a data rate and robustness comparable to AquaScope. We integrate AquaApp systems with the generative image codec to make it comparable to AquaScope.
- *CSS+ [89]:* We also implement a conventional CSS-based acoustic system based on [89] as another baseline system. In contrast to AquaScope, CSS+ adds 4 training symbol chirps (2 up-chirps and 2 down-chirps) at the beginning of the packet and leverages FFT bin offset to perform symbol time synchronization [89]. CSS does not perform channel equalization. Similarly, we integrate CSS with generative image codecs proposed by AquaScope for fair comparison.
- *PNG/JPEG:* We consider two traditional image codecs, PNG and JPEG, to compare with generative codecs with various scales and quality levels, respectively. We implement those codecs on AquaScope.
- *Classic NN [20]:* We also compare our generative codec with a classic neural codec that employs entropy coding with learned probability models for image compression.

We consider two primary metrics when evaluating AquaScope and baseline schemes (unless otherwise stated).

- *Index Error Rate (IER):* First, we use IER, defined as the the number of corrupted tokens over the all tokens of the image during transmission, as one performance metric. Image quality is directly determined by the correctness of indices, and IER also reflects the effectiveness of the techniques at the PHY layer.
- *LPIPS:* Second, we use *LPIPS* [99] as a perceptual metric to measure image fidelity by comparing deep feature representations of the received and reference images. To evaluate the effectiveness of LPIPS, we manually score the received images with reference to the original image, categorizing them as cannot understand, moderate understanding, and well understanding. We then calculate correlations with the following image metrics: PSNR, SSIM [33], MS-SSIM [85], VIF [73], LPIPS [99], CLIP score [66], and DinoV2 score [61]. CLIP and DinoV2 scores are cosine similarities between embeddings of received and ground truth images, calculated by large vision models known for semantic representation extraction. Among all the metrics, LPIPS shows the highest correlation, at −0.4. Hence, we use LPIPS as our image metric.

6.2 Evaluation Results in the Wild

End-to-end performance. First, we present the end-to-end performance of AquaScope compared to baseline schemes

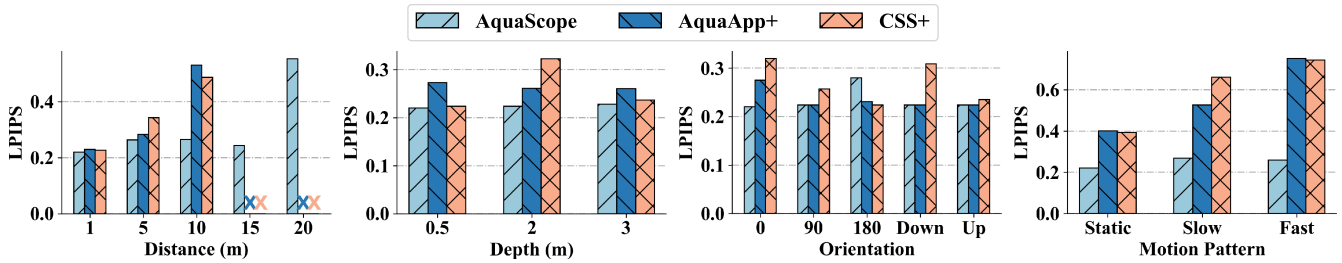


Figure 17: Impact of different physical factors. An "X" mark indicates that the receiver does not receive the image packets.

in real underwater scenarios. Fig. 15 illustrates compression efficiency, i.e., the tradeoff between image quality (in LPIPS) and data size (in bits) across various schemes. Specifically, we use three instances of PNG at 10%, 20%, and 30% scale ratios (PNG-10%, PNG-20%, and PNG-30%), JPEG at quality level 10 (JPEG-Q10), and a neural codec (NN) at quality levels 1 and 3 (NN-Q1 and NN-Q3). Higher JPEG/NN quality levels or larger PNG scale ratios result in significantly larger data sizes, which are impractical for current underwater mobile systems.

From the figure, we can see that AquaScope consistently achieves the best compression efficiency across all baselines. In terms of transmission protocols, AquaScope achieves 36.7% ∼ 40% performance gains in LPIPS compared to AquaApp+ and CSS+. The proposed interleaved training symbols achieve better time synchronization performance than CSS+. Using FFT bin offset for time synchronization in CSS+ only works for specialized hardware that operates with much higher transmission power. AquaApp+ suffers from transmission errors due to its suboptimal carrier selection for long packets and thus has larger LPIPS. Notably, AquaScope, CSS+, and AquaAPP+ all adopt generative codecs, where data size remains fixed. For baseline codecs, we report results only in error-free cases, as they fail to recover images once errors occur. In contrast, AquaScope integrates results across both error-free and error-prone cases (10.5% IER in average), yet still achieves superior compression efficiency and strong robustness under transmission errors due to its generative reconstruction capability. The decoded image samples can be seen in Fig. 16, and more qualitative results can be found in Fig. 25 and Fig. 24 in the Appendix (§10).

Effect of distance. The first image in Fig. 17 plots LPIPS versus distance. Several observations can be made. When the sender–receiver distance is short (<5m), all schemes maintain acceptable visual quality with IER below 11% (2% BER); notably, AquaScope incurs no index errors. At medium distances (10–15m), AquaScope still maintains low LPIPS, while CSS+ and AquaAPP+ degrade sharply to 0.49 and 0.53, with IER rising to 31% (6% BER) and 46% (19% BER), respectively. In this case, the recovered images are indistinguishable compared with the original images. At 20m, LPIPS for AquaScope begins to increase. The mark "X" in the figure indicates that the image packets are not received at the receiver side. Based on the result, AquaScope can reliably deliver images with a distance of up to 20m, and the performance starts to decrease when the distance is larger than 20m due to signal attenuation.

Effect of depth and orientation. The second and third images in Fig. 17 show LPIPS versus depth and orientation, respectively. For depth, we placed the mobile phones close to the water surface, at the middle (2m) and the bottom (3m) of the lake, respectively, with a distance of 5m. For orientation, we first set the speaker of the sender and the microphone of the receiver to face each other. Then, we rotate the receiver phone in the azimuth angle from 0° to 90° and 180°, respectively. Furthermore, we also rotate the pitch angle of the receiver phone to 90° and −90° (named up and down) to test the effect when the speaker and microphone are not in the same plane. We mount the phone to a phone holder connected to a selfie stick and submerge the phone underwater to about 1.5m depth with our hands holding the stick. The results are shown in Fig. 17. We can see that depth and orientation changes of the mobile phones have less impact on the systems, and LPIPS is generally below 0.3 (15% IER, 2% BER). The variance in the result may come from the slight movement during the experiment.

Effect of mobility. The last image in Fig. 17 shows LPIPS versus different mobility at the distance of 5m. We submerge two mobile phones connected to tape measures from the bridge in the lake and keep the sender still. We randomly move the receiver by moving one end of the tape measure moderately and intensively. As depicted in the figure, AquaScope shows great resilience to the mobility and maintains LPIPS within 0.3 (IER within 8%, BER within 1.5%) while LPIPS of CSS+ and AquaApp+ is over 0.4 (50% indices are wrong) when the mobile devices move.

6.3 Deep Dive

Fine-tuning. To illustrate the advantage of error-resilient fine-tuning, we consider two new baselines: TiTok [97] and AquaScope-WE. TiTok is the state-of-the-art generative model trained on ImageNet [68] and AquaScope-WE is fine-tuned on our collected underwater image dataset without introducing transmission errors. We simulate the IER from 0% to 20%

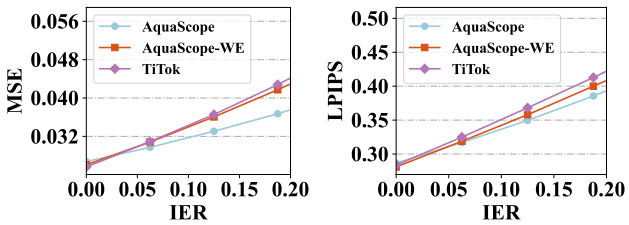


Figure 18: Impact of fine-tuning on image quality.

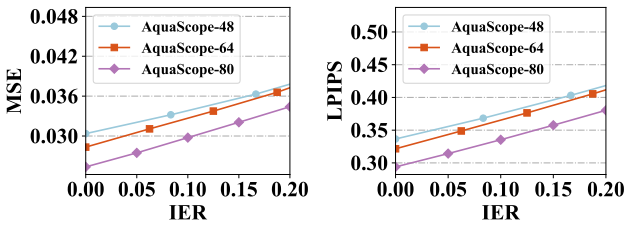


Figure 19: Impact of token number M on image quality.

and the results are shown in Fig. 18. We find that as the IER increases, the LPIPS score and mean square error (MSE) of all approaches also rise. Furthermore, AquaScope can achieve up to 8% and 21% gains in the LPIPS and MSE, respectively, compared to the baselines. This demonstrates AquaScope is more robust to the transmission errors after fine-tuning. The visualization examples can be seen in Fig. 23 in Appendix.

Token numbers. We train different transformer versions with 48, 64, and 80 tokens from scratch, namely AquaScope-48, AquaScope-64, and AquaScope-80, respectively. The results are shown in Fig. 19. AquaScope-80 has better LPIPS and MSE as it compresses the image with more (redundant) tokens and thus is more robust to the index errors. To balance robustness and computation, and communication overhead, we choose $M = 64$ tokens.

Time equalization and synchronization. We examine the importance of reliability enhancement techniques at the PHY layer proposed by AquaScope, as illustrated in Fig. 20. The SER is used as the evaluation metric because it accurately reflects errors (before channel decoding) in the chirp-based communication system. When only a single equalization is applied at the beginning of the entire image packet (referred to as "one Equal."), the system exhibits an average SER exceeding 50%, indicating that a single initial equalization is insufficient to account for the evolving channel conditions in packets lasting several seconds. Introducing multiple training symbols for repeated equalization, but without employing symbol time synchronization (referred to as "wo Sync."), reduces the average SER to below 20%. Utilizing cross-correlation to estimate the start time of each training symbol, but without applying the smoothed and bounded time offset correction (referred to as "wo S+B"), mitigates the time offsets caused by movement, further lowering the SER to 11%. Finally, applying the smoothed and bounded time offset correction, as described in Sec. 4.4, reduces the SER by an additional 2%, sometimes correcting up to five additional index errors.

6.4 System-level Performance

Latency breakdown. The averaged end-to-end latency of AquaScope is 9.2s with a standard deviation (std) of 0.66s. The average latency of the AquaApp+ is 10.9s and the std is 3.0s. Fig. 21 shows latency breakdown of AquaScope and AquaApp+ that contains image encoding, signal generation including channel coding and modulation, transmission, signal recovery including demodulation and channel decoding, and image decoding latency. We found transmission latency is the bottleneck of the end-to-end system latency. Compared to AquaScope, AquaApp+ has much more variations in transmission latency as its data rate changes according to varying channel quality.

Table 1: Resource Usage

Device	CPU	Memory
Sender	Img. Enc. (55%) Sig. Gen. (43%)	2 GB
Receiver	Img. Dec. (46%) Sig. Rec. (15%)	2.5 GB

Energy. Fig. 22 illustrates the battery percentage drained at both the sender and receiver over approximately 50 minutes of continuous image transmission. The battery has a capacity of 4,000 mAh and is in normal condition. The results indicate that the battery level decreases by only 20%, suggesting that the device could last for over three hours on a full charge. In contrast, recreational diving, typically limited by the size of the gas tank, lasts for up to an hour [11], which is significantly shorter than the battery life of a mobile phone when using AquaScope. Hence, AquaScope is practical for recreational diving and can also support professional diving which lasts for a longer time.

Resource Usage. We also evaluate the resource usage of mobile devices when running the generative models and signal processing pipeline. As shown in Table 1, the optimized TiTok model incurs approximately 55% CPU utilization (8-core) for image encoding and 46% for decoding. For signal processing, signal generation and reconstruction consume about 43% and 15% CPU, respectively. Memory usage on the sender and receiver is approximately 2.0 GB and 2.5 GB, respectively. The receiver requires more memory because upsampling generates a larger volume of intermediate activations.

7 Discussion

In this section, we discuss the potential improvements of AquaScope.

Non-line-of-sight. Currently, our system experiences performance drops when the sender and receiver are in a non-line-of-sight condition, primarily due to severe multipath effects and attenuation. The obstruction can be caused by

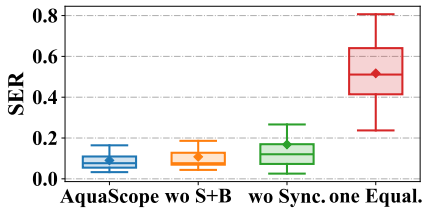


Figure 20: Ablation results at PHY.

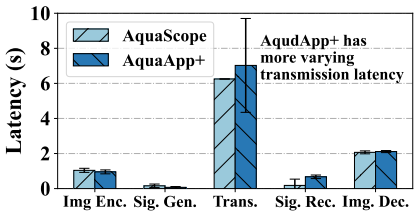


Figure 21: Latency breakdown.

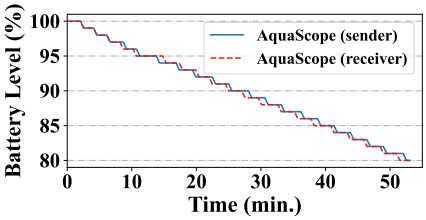


Figure 22: Battery drain over time.

the human body (whether it involves the sender, receiver, or both) or by large obstacles like coral or shipwrecks that completely block the line of sight between the sender and receiver. For the first type of obstruction, one possible solution is that the sender and receiver (upon hearing the transmission sound) can opt to expose the mobile phone to reduce the obstruction. The second type of obstruction, involving large obstacles, is uncommon and currently not supported by our system.

Interference from pouch. Although chirp signals can resist interference and noise, they fail under wideband interference at low SNR. In our experiment, the soft pouch generated such interference when it deforms due to movement. A practical solution is to use a hard case instead of a soft pouch with extra attention brought by the hard case. Alternatively, filling the pouch with more air can help resist shape changes and reduce interference.

End-to-end latency. Based on the latency breakdown discussed in §6.4, several methods can be implemented to reduce the end-to-end delay. First, image encoding and decoding currently run on CPU which is time-consuming but they can be migrated to on-device GPUs to speed up the inference with Vulkan backend [2]. Second, we can hide most of the processing latency via pipelining and only the transmission latency becomes the bottleneck.

Preamble detection. Current preamble design and detection algorithm is vulnerable to interference and can be further improved. In the future, more advanced techniques such as neural network-based preamble design and detection [92], can be leveraged to improve the packet detection reliability.

Underwater factors. AquaScope currently does not consider underwater factors such as water current and temperature. Future work can explore more adaptive chirp transmission and synchronization mechanisms that explicitly account for these environmental conditions.

8 Related Work

In this section, we discuss additional related work beyond the underwater image compression techniques and specialized acoustic hardware introduced in §2.

8.1 Underwater Communication

Mobile devices. AquaApp [15] is the first work that enables underwater text message transmission between two mobile phones. They further extend the system for 3D localization underwater [16]. AquaHelper [91, 92] focuses on designing robust signal detection algorithms for SOS message transmission. The information transmitted by their systems is very limited. Although some work in [50, 51] attempts to communicate via QR codes on mobile devices. Successfully scanning the QR code heavily depends on the underwater visibility and the distance between the sender and receiver. Instead, AquaScope is capable of transmitting images on mobile devices.

Light-based communication. Beyond acoustics, prior work has explored using optical signals to communicate underwater. Amphilight [13] enables low-power communication by modulating customized LED light sources to transmit information through water, while Shrimp [47] leverages laser diodes to achieve higher-directionality optical links. However, the optics-based communication system is mainly limited by the transmission distance and the light condition underwater.

Chirp-based acoustic systems. Chirp-based transmission protocol is widely adopted for underwater communication [39, 67, 75, 76, 78, 88]. They mainly focus on the optimization of the transmission protocols such as rate adaptation, modulation designs, multi-path elimination. Their designed protocols still are for a small packet and require larger transmission power and bandwidth which cannot be directly applied for image transmission on mobile devices. AquaScope optimizes the chirp-based protocol tailored for robust image transmission on commercial mobile devices.

8.2 Generative Compression

Generative compression has attracted significant attention [8, 14, 21, 29, 43, 46, 52, 87] and has been successfully applied to image/video codecs and wireless communication, where it reduces communication overhead without compromising the quality of the received images or videos. AquaScope is the first system to apply generative compression to underwater communications, and it also extends this capability to other limited-bandwidth or error-prone scenarios—such as satellite networks and rural communication.

Image tokenization [74, 90], as a way for generative compression, learns a compact image representation for various downstream tasks such as classification, reconstruction, and generation. Since the introduction of neural discrete representation [81], numerous methods [23, 94, 95, 97] have

been developed. Among these, VQGAN [23] and TiTok [97] were selected for our system due to their balance between compressed image size and decoded image quality.

9 Conclusion

We present AquaScope, the first underwater acoustic system that enables image transmission on mobile devices. By combining generative image compression with reliability-enhancing techniques at the PHY layer, AquaScope transmits 256×256 color images with low latency across various underwater conditions, preserving high perceptual quality even under stringent bandwidth limitations and severe transmission errors. By extending underwater communication from text messaging to image transmission, AquaScope contributes to the ongoing development of underwater technologies.

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Figure 23: Samples of reconstructed images from different methods. VQGAN uses 256 tokens. TiTok employs the base-64 model with a 20% IER. AquaScope-WE is our fine-tuned version of TiTok model without transmission errors. AquaScope represents our fine-tuned version of TiTok with a 20% IER.

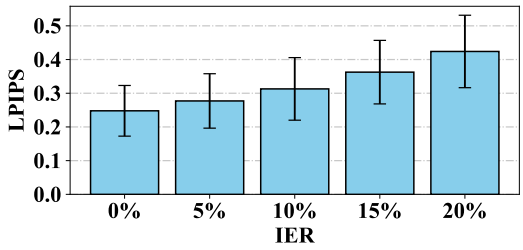


Figure 24: Impact of IER on AquaScope.

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10 Appendix

10.1 Visualization examples.

We illustrate the reconstructed images from different versions of generative models in Fig. 23. Our first observation is that the image reconstructed from TiTok exhibits similar or even better visual quality than the one reconstructed using the VQGAN model even with less token numbers. Our second observation is that when the perturbed tokens are identical—where we control both the perturbed tokens and their values to be the same in the last two images—our model, leveraging the recovery capability of error-resilient fine-tuning, retains many visual details. In contrast, the baseline images lose both the visual details and the semantic meaning of the original image.

10.2 Qualitative results of generative compression

In this part, we further demonstrate the performance and generalizability of generative codecs on a self-collected dataset containing 400 images sourced from 10 YouTube videos. These videos were taken from 10 distinct locations to cover a variety of underwater scenes, including lakes and oceans, featuring underwater creatures, divers, caves, and shipwrecks. The 400 images were manually selected from a pool of 1000 images to eliminate duplicates and blurred images. The pre-trained TiTok model is used in a zero-shot manner for compression, with the dataset serving solely as input and not being used for fine-tuning or any task-specific training. Representative samples and qualitative results of generative compression are shown in Fig. 25.

Generative compression works well across a variety of images, differing in objects, colors, and complexity. The recovered images, when no errors are present (as shown in the second row), preserve most of the semantic meaning and details. Even with a 10% IER (where approximately 6-7 indices out of 64 are randomly altered), the images largely retain their original semantic meaning with an LPIPS score lower than 0.25 in most cases. However, the semantic meaning of the received images begins to degrade when the IER exceeds 15%. Interestingly, some images, such as those in the second and last columns, are more resilient to errors. This suggests that even with a high IER, the original semantic meaning may still be perceptible in certain cases. Another notable property is that errors in the received tokens do not alter the semantic meaning of the recovered image. As shown in the last row, images with a high error rate lose their semantic meaning entirely. This ensures that the receiver will either perceive the correct semantic meaning or none at all, but will not interpret an incorrect meaning. The statistical results are shown in Fig. 24.

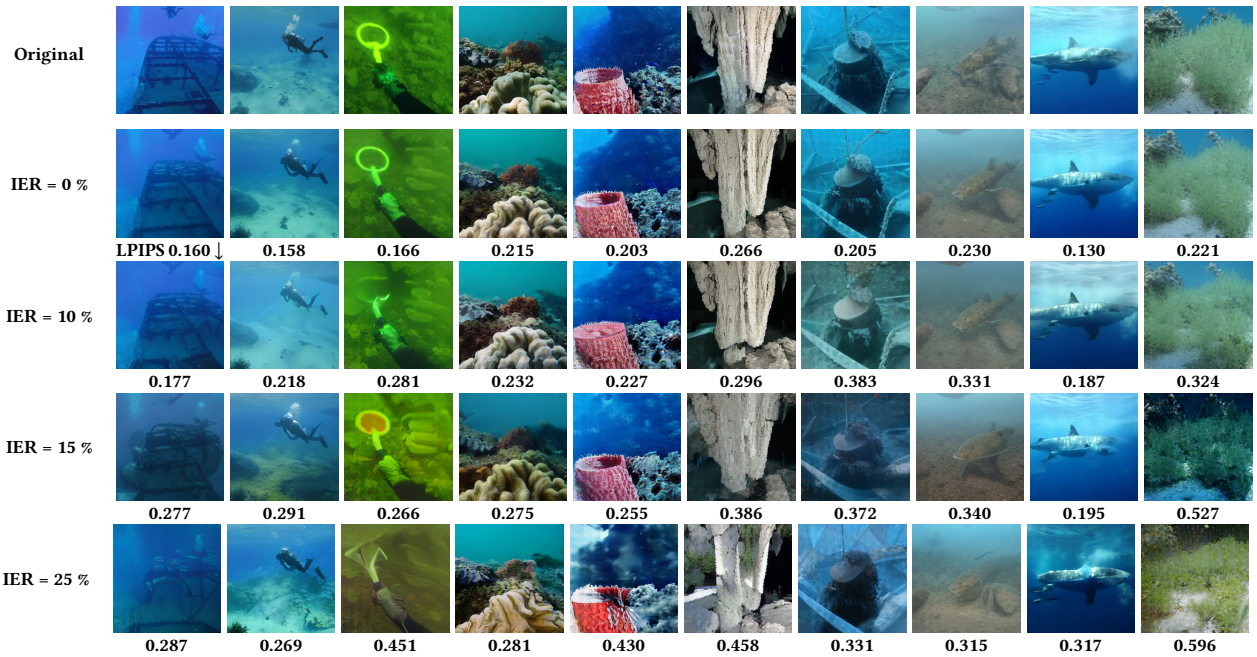


Figure 25: Qualitative results of generative compression are presented. From left to right, the columns depict images of a shipwreck, a scuba diver, a treasure hunt event, coral, stalactites in underwater caves, fish, and plants. The top row shows the origin image taken by the camera and the other rows show the decoded image under different IER. The caption for each image is its LPIPS (the lower the better) with the origin image as the reference.